DISCRETE MATH · COUNTING AND PROBABILITY

Counting Outcomes Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Counting Outcomes Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. How many 2-letter "words" can be made from A, B, C, D if letters can repeat? If they cannot?

2. Is choosing a 2-person committee from 4 people the same count as choosing a president and vice-president? Predict both.

2 Explore use the tool

1. Order matters, no repetition: find n and r with 60 outcomes.

2. Selections of 2 from 5: group the orderings. How many orderings per selection? How many in all?

3. Selections from 5: compare r = 2 and r = 3. What do you notice?

4. 3 scoops from 4 flavors, repeats allowed, order ignored: record the count and two outcomes.

3 Record use the tool and fill in the table

Four counting models

n = 5 letters, r = 3 chosen.

Order matters?	Repetition?	Formula	Count
yes	yes		
yes	no		
no	no		
no	yes		

Probability by counting

Situation	Favorable	Total	Probability
Arrange A, B, C, D: starts with A			
2 letters from A–F with repetition: at least one A			
Choose 2 of A–E: the pair contains A			

4 Explain what did you discover?

1. P(n, r) = C(n, r) · _____ because each selection can be ordered in r! ways.
2. C(n, r) = C(n, _____).
3. P(E) = favorable/total only when all outcomes are _____ likely.
4. Roll two fair dice. A student counts 21 unordered results and says P(double 6) = 1/21. Explain the error and give the correct probability.

5 Apply a new situation, no tool

1. How many 4-digit PINs have no repeated digit?
2. A club of 12 chooses a 3-person committee. How many committees?
3. How many ways can 8 runners finish 1st, 2nd and 3rd?
4. Flip 5 coins. Find P(exactly 2 heads).